Strathmore
UNIVERSITY

SCHOOL OF COMPUTING AND ENGINEERING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATICS

END OF SEMESTER EXAMINATION

ICS 1201 OBJECT-ORIENTED PROGRAMMING I

DATE: 7^{th} March 2024 Time: 08:00-10:00 Hours

Instructions:

1. This examination consists of **FIVE** questions.
2. Answer **Question ONE (COMPULSORY)** and any other **TWO** questions.

QUESTION ONE (COMPULSORY) (30 MARKS)

A. Java is a suitable tool for studying OOP; explain **any two** features of Java language that justifies it for this field of study **(2 marks)**

B. In OOP explain the use of the following keywords **(3 marks)**

- ***i.*** ***final***
- ***ii.*** ***static***
- ***iii.*** ***super***

C. Java doesn't allow *multiple inheritance* instead it has **interfaces** and **abstract** classes. Explain any 3 differences between an *interface* and *abstract class* **(6 marks)**

D. A java beginner has had of the statement "java is *compiled* then *interpreted*" Explain to them why this claim is true **(2 marks)**

E. You have been hired as a new developer to convert an old App that runs on Structured Programming to run on Object Oriented architecture.

- *i.* Explain the meaning of the FOUR OOP principles giving practical examples demonstrating their purpose of how you will use them in a code example. **(8 Marks)**

ii. Explain why you may need a java API and give one example of an API you may use stating its purpose **(4 Marks)**

F. Draw the UML Design Diagram from which the below Java Code was written **(5 marks)**

```java
public class GradeBook
{
   private String courseName; // course name for this GradeBook

   // method to set the course name
   public void setCourseName( String name )
   {
      courseName = name; // store the course name
   } // end method setCourseName

   // method to retrieve the course name
   public String getCourseName()
   {
      return courseName;
   } // end method getCourseName

   // display a welcome message to the GradeBook user
   public void displayMessage()
   {
      // calls getCourseName to get the name of
      // the course this GradeBook represents
      System.out.printf( "Welcome to the grade book for\n%s!\n",
         getCourseName() );
   } // end method displayMessage
} // end class GradeBook
```

QUESTION TWO (15 MARKS)

A. Write the Java code for **class Account** as shown in the UML. **(8 Marks)**

Account
-accountNumber : string -balance : double
+Account(in accNo : string, in bal : double) +setBalance(in balance : double) : void *+getBalance() : double* +showDetails()

B. Implement ***showDetails()*** method such that it displays the account number and balance values. **(3 Marks)**

C. Using a Java code excerpt, write a **Main class** that declares **2 objects** of class Account. The objects should display the account number and balance. **(4 Marks)**

QUESTION THREE (15 MARKS)

A. You wish to write OOP software for a Watch manufacturing company. Write java code that will declare the class shown in the below UML model. The **getAge()** method should calculate and return the present age of a watch. *Hint:* Use yom (year of manufacture). The **showDetails()** method should display the full name of the watch together with the type and age in separate lines. All other methods can remain empty. **(10 marks)**

Watch
-name: String -type: String -yom: int
+Watch(name: String) +Watch(name: String, type: String, yom: int) +setType(type: String): void +setYom(yom: int): void +getName(): String +getType(): String +getAge(): int +showDetails(): void

B. List and **Explain** TWO types of errors that a Java programmer might encounter while writing and compiling a java code. **(4 Marks)**

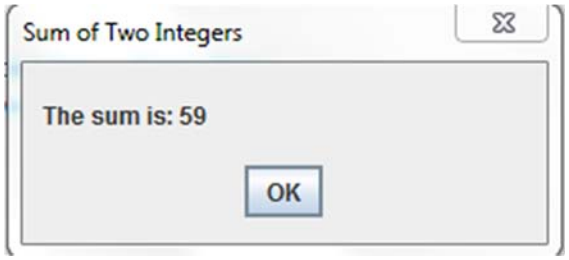
C. Explain what **exception handling** is as used in Java programming. **(1 Mark)**

QUESTION FOUR (15 MARKS)

A. In Java GUI design, explain what a JOptionPane is **(2 Marks)**

B. Give at least 3 scenarios where a Java programmer may use a JOptionPane. **(6 Marks)**

C. Write a Java code excerpt that will produce the output shown in the figure below. **(7 Marks)**



QUESTION FIVE (15 MARKS)

A. Differentiate between an **abstract class** and a **concrete class**. **(4 Marks)**

B. Describe what a **static method** is. Give one circumstance where Java uses static methods in its applications. **(4 Marks)**

C. An abstract class *Triangle* has *sideA*, *sideB* and *sideC* as attributes. It has an abstract *getPerimeter()* method. Declare a concrete class *RightTriangle* to inherit the class Triangle. **(7 Marks)**